Marking Scheme and Rubric

Machine Learning for Business (QBUS6810)

1 Assessment overview

The report and code are assessed together for accuracy, reasoning, supporting evidence and clarity. Each subquestion receives one overall mark up to the maximum shown. The question numbers match the assignment instructions.

Component	Marks
Business objective	5
Data understanding and representation	22
Model development	30
Final evaluation	20
Deployment and operations	10
Communication and reproducibility	5
AI use and reflection	8
Total	**100**

Communication is assessed across the report. AI reflection comprises four cases worth 2 marks each (8 marks in total).

2 Assessment criteria

Group: _____ Marker: _____

1. Business objective (5 marks)

Question / assessment criteria	Max	Mark
1a. Business problem and purpose. Clear explanation of the client's business problem, who will use the predictions, and the practical value the project could provide.	5	_____

Subtotal: _____ / 5

2. Data understanding and representation (22 marks)

Question / assessment criteria	Max	Mark
2a. Data structure and join. A verified join, a correct distinction between duplicate records and separate sales, and justified treatment of repeated records.	4	_____
2b. Data quality. Identification of relevant predictor-quality issues and evidence-based treatment decisions.	4	_____
2c. Exploration. Clear evidence of temporal and geographic price patterns, reasoned explanations and links to modelling choices.	6	_____
2d. Feature engineering. Thoughtful feature engineering and representation suited to the data and models.	4	_____
2e. Availability and sample limits. Awareness of leakage and sample-selection risks, with a clear interpretation of what is unobserved for unsold homes.	4	_____

Subtotal: _____ / 22

3. Model development (30 marks)

Question / assessment criteria	Max	Mark
3a. Baselines. A meaningful baseline and a clear explanation of its role in evaluation.	3	_____
3b. Model choices. Correct implementation, reasoned model choices and fair comparisons that account for the available data.	5	_____
3c. Validation strategy. A justified validation strategy that considers time and property overlap, with sound separation of model development and final evaluation.	7	_____
3d. Experiments and learning. Systematic experiments, fair feature-set comparisons and evidence of learning from results.	8	_____
3e. Tuning and complexity. Reasoned tuning and control of model complexity, supported by evaluation evidence.	5	_____
3f. Final selection. Clear final selection supported by the model comparison and valuation objective.	2	_____

Subtotal: _____ / 30

4. Final evaluation (20 marks)

Question / assessment criteria	Max	Mark
4a. Held-out performance. Appropriate regression metrics with clear justification. Correct evaluation and interpretation of price errors, residuals and baseline comparisons.	6	_____
4b. Time, location and errors. Meaningful comparisons and evidence-based explanations, with attention to sample sizes and the limits of the evidence.	5	_____
4c. Error priorities. A justified choice of error weights, correct weighted-error calculations, a meaningful baseline comparison and clear interpretation of the alternative scenarios.	7	_____
4d. Recommendation. A recommendation for or against a pilot supported by the evaluation and its limitations.	2	_____

Subtotal: _____ / 20

5. Deployment and operations (10 marks)

Question / assessment criteria	Max	Mark
5a. Practical use. A workable process with relevant input requirements, review and use of the estimate.	2	_____
5b. Monitoring and response. Relevant monitoring measures linked to specific findings and practical responses.	3	_____
5c. Risks and further evidence. Two relevant risks or constraints, their implications and feasible responses grounded in the chosen model and setting.	3	_____
5d. Valuation review case. Sound judgement about the new evidence and a justified approach to reviewing the estimate.	2	_____

Subtotal: _____ / 10

6. Communication and reproducibility (5 marks)

Question / assessment criteria	Max	Mark
6a. Report quality. Uses a clear structure and readable exhibits. Explains claims precisely and provides the necessary references.	2	_____
6b. Reproducible code. Code runs in a clear order, uses portable paths, records relevant settings and packages, and reproduces the reported results.	3	_____

Subtotal: _____ / 5

7. AI use and reflection (8 marks)

Each case receives one overall mark out of 2, based on all four points:

1. Clear record of the initial view, prompt and relevant AI advice.
2. Independent verification supported by evidence from the analysis or reliable sources.
3. A reasoned decision to accept, change or reject the advice, and its effect on the work.
4. Specific learning and remaining limitations.

Full marks require all four points to be addressed clearly and supported by evidence. Partial marks reflect gaps in coverage, verification or reasoning. An absent case receives zero.

Question / assessment criteria	Max	Mark
7a. Data or features.	2	_____
7b. Validation or leakage.	2	_____
7c. Modelling or debugging.	2	_____
7d. Business interpretation.	2	_____

Subtotal: _____ / 8 **Total:** _____/100